

No. of Functional Features Included in the Solution

Date	1 July 2026
Team ID	SWTID-2026-9706
Project Name	Voice Based Concept understanding Analyser
Maximum Marks	5 Marks

Functional Features Overview:

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Audio Upload & Playback	Upload WAV/MP3 and play it back in-browser	app.py	Done	—
2	Speech-to-Text Transcription	Converts audio to text using Whisper	speech_to_text.py	Done	—
3	Semantic Similarity Scoring	Compares transcript to reference concept via Sentence-BERT	semantic_eval.py	Done	—
4	Audio Feature Extraction	Extracts pause ratio and RMS energy via Librosa	audio_utils.py	Done	—
5	Scoring Engine	Combines all metrics into final score & understanding level	scoring_engine.py	Done	—
6	Filler Word Detection	Calculates filler word ratio from transcript	audio_utils.py	Done	—
7	Waveform Visualization	Generates and displays waveform plot	audio_utils.py / app.py	Done	—
8	PDF Report Generation	Downloadable PDF with transcript, waveform, evaluation summary	report_generator.py	Done	—

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Audio upload & playback, results dashboard
2	Backend / Logic	4	Speech-to-text, semantic similarity, audio features, scoring engine
3	Database / Storage	1	Session state (in-memory, no persistent DB by design)
4	API / Integration	1	Whisper & Sentence-BERT models (local, no external API calls)
5	Security / Authentication	0	Not applicable — no login required by design